

Josko de Boer — Freelance Data Engineer

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Freelance Data Engineer with proven expertise in building and leading teams across AWS, GCP, and Azure environments. Strong focus on end-to-end data engineering solutions, software development, and process automation to enhance team efficiency. GenAI enthusiast.

- Developed and deployed scalable data products on AWS for Randstad and Alliander, GCP for Azerion and Spil Games, and Azure for Nationale Nederlanden, leading to operational efficiency improvements.
- Spearheaded Azure+Databricks-based data platform management for Nationale Nederlanden, streamlining workflows and integrating CI/CD pipelines.

As a former consultant, I am used to onboarding rapidly and ensuring high-quality delivery.

Skills

Programming	Python, SQL, C++14, Java, PHP, Javascript, Fortran2000, Labview, MatLab, Maple, Bash, HTML, CSS, Terraform	GenAI	Athena, Redshift, Data Vault, Star Models
Tools & Libraries	Numpy, Scipy, Pandas, PyMC3, FastApi, jQuery, Jupyter, AWS SDK, GCP SDK, Matplotlib, Seaborn, requests, scikit-learn, SQLAlchemy, unittest, pytest, Airflow, networkx, Makefiles, Kafka, Databricks, Polars	GCP	Open WebUI, Ollama, Dspy
		AWS	Pub/Sub, Dataflow, Kubernetes, BigQuery, DataStore, Composer, Cloud Build
Databases	BigQuery, MySQL, PostgreSQL, RDS, Aurora,		CloudFormation, ECS, Batch, Kinesis, LoadBalancer, Redshift, S3, Lambdas, Cloud Development Kit (CDK), API Gateway, Athena, RDS
		Azure	Storage (ADLS Gen 2), Network, Data Factory, Virtual Machines, Databricks

Experience

Xomnia

Feb 2021 - Dec 2024

Senior Data Engineer

Client: Nationale Nederlanden (2024-2025)

- Engineered Azure-based data platforms using Data Factory and Databricks, scaling to support 10+ teams with Medallion architecture.
- Developed a long-term architecture plan, refactoring Terraform scripts to optimize infrastructure and workflows.
- Directed the integration of CI/CD pipelines, reducing release cycles and enhancing deployment efficiency.
- Created prototypes for incorporating streaming data into their Data Lakehouse

Internal Role: Head of Data Engineering (2022-2024)

- Managed a team of 20 data engineers, overseeing HR, sales, and team satisfaction initiatives from 2022-2024.
- Created business proposals for team expansion and guided Learning & Development (L&D) programs.
- Worked on internal tooling to develop genAI/LLM proposition using openAI API (ChatGPT), and customized hosting using Open WebUI, Ollama and Dspy.
- Coaching and Development for the team, on GCP, Azure and AWS
- Setup model architectures for various clients.

Client: Alliander #3 (2023-2024)

- Developed simulation models for electrical grid future needs using AWS Batch and Polars, executing 35,000+ simulations.
- Orchestrated infrastructure planning and DevOps enhancements, managing the migration of the python/SQL code to Polars to optimize data processing.

Client: Alliander #2 (2023)

- Designed a greenfield data platform integrating in-house tools to streamline workflows for specialists.
- Directed the development of both backend API and frontend interfaces with long-term architecture planning for scalability.
- Created and gave courses in various topics on Data and ML engineering, both on GCP and Azure

Client: Alliander #1 (2022-2023)

- Engineered a data platform for long-term electricity network calculations, implementing comprehensive cost and quality control measures.
- Built an AWS-based platform using AWS Batch (Python), RDS, API Gateway, PostgreSQL, and GIS enhancements to deliver a full API for the backend while hosting the frontend.
- Optimized retrieval SQL operating on SAP HANA, Snowflake and Redshift

Client: Randstad NL (2021-2022)

- Enhanced the data platform by implementing Infrastructure as Code (IaC) and DevOps best practices.
- Developed and deployed an MLOps workflow to improve machine learning pipeline management.
- Refactored and optimized data pipelines for better performance and maintainability.

Internal Role

- Coordinated the 2021-2022 traineeship program, overseeing training modules, and mentoring new engineers.
- Optimized existing SQL and Python internally.
- Joined in setting up model architectures for various clients.

Spil Games / Azerion

2019 - 2021

Head of Data Engineering

- Onboarded new teams into the data lakehouse architecture on GCP post-Azerion acquisition.
- Developed unified streaming pipelines to support various game categories, facilitating scalable data ingestion and processing.
- Executed financial planning products for each game and orchestrated architecture planning for scalable, long-term infrastructure solutions.
- Started migration to AWS, creating a hybrid cloud environment
- Expanded use Kubernetes and Python in data tooling

Spil Games

2018 - 2019

Data Engineer

- Built and optimized data lakehouse solutions using GCP technologies (Composer, Datastore, Pub/Sub, Dataflow, BigQuery (SQL)) and custom code (Python)
- Implemented cost-effective data lineage tools and enhanced recommender systems, reducing data processing costs and improving the overall architecture.

Jan Zaanen Group

2017 - 2018

Ph.D. Student

- Taught multiple master's courses and conducted research on high-Reynolds number hydro simulations within the AdS/CMT framework.

Nynke Dekker Lab

Sept 2016 - 2017

Technician: Scientific Programming and Analysis

- Developed software for scientific setups using LabView, and automated data analysis workflows with Python and Matlab.
- Conducted critical reviews of data analysis methods to ensure accuracy and reliability in research outputs.

Education

TU Delft

M. Sc. Quantum nanotechnology

B. Sc. Applied Physics

2009-2016

Casimir Pre-PhD programme

Minor in Education